



Rewind: A Past-Self Intervention via Character-Mediated Regret Replay

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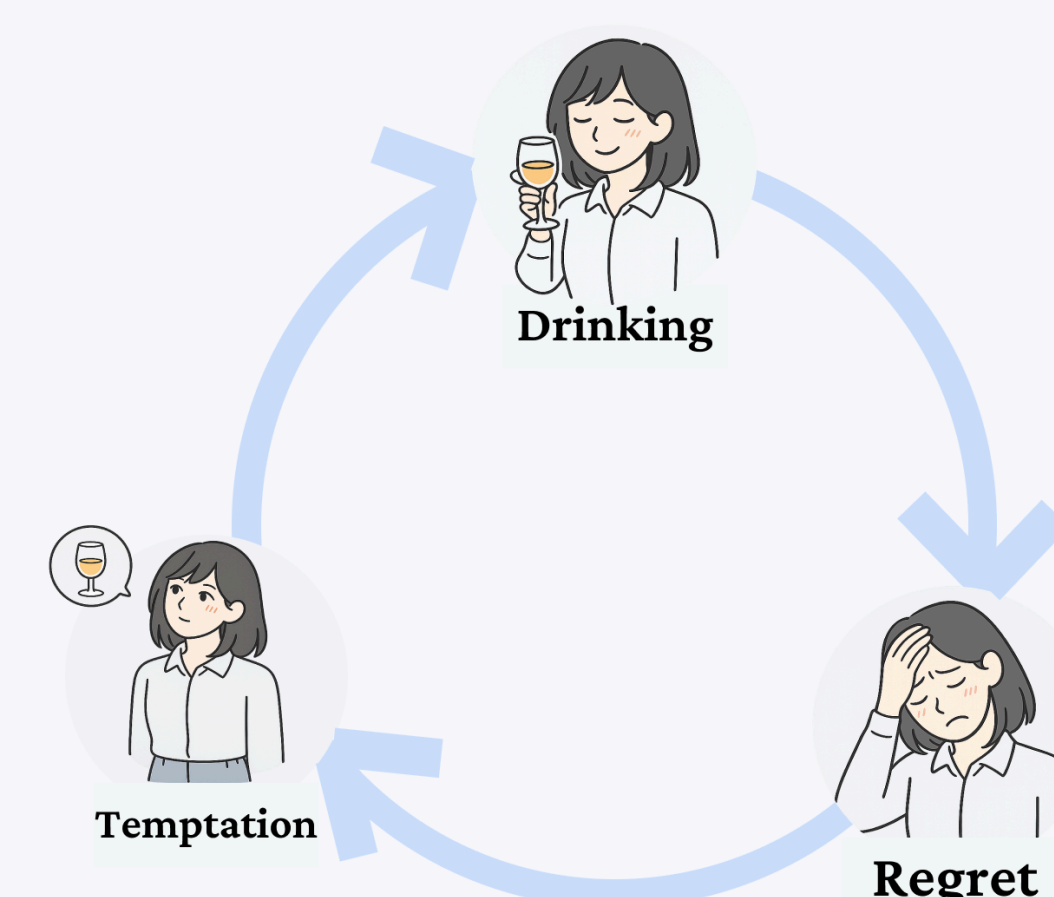
Figure 1. Example use scenario of **Rewind**. After a late-night scrolling episode, the user records a regretful reflection. When the same temptation returns the next night, the recorded past self is replayed through the cat avatar to create a pause for reconsideration, leading to a better-rested next morning.

Abstract

Rewind is a concept for helping people interrupt repeated unwanted habits at the moment they are most likely to happen. After a regretful episode—such as staying up too late scrolling on the phone—the user records a short reflection with voice and facial expression. The system then turns that moment into a re-playable clip through a cat avatar that mirrors the user’s emotion in a softer, less threatening way.

When a similar temptation returns later, the user can replay this past-self message to interrupt the usual habit loop and create a brief pause before acting. Rather than treating regret as a feeling that simply fades away, Rewind explores whether regret can be captured and brought back at the right time in a more approachable form.

Without Past-Self Intervention



With Past-Self Intervention

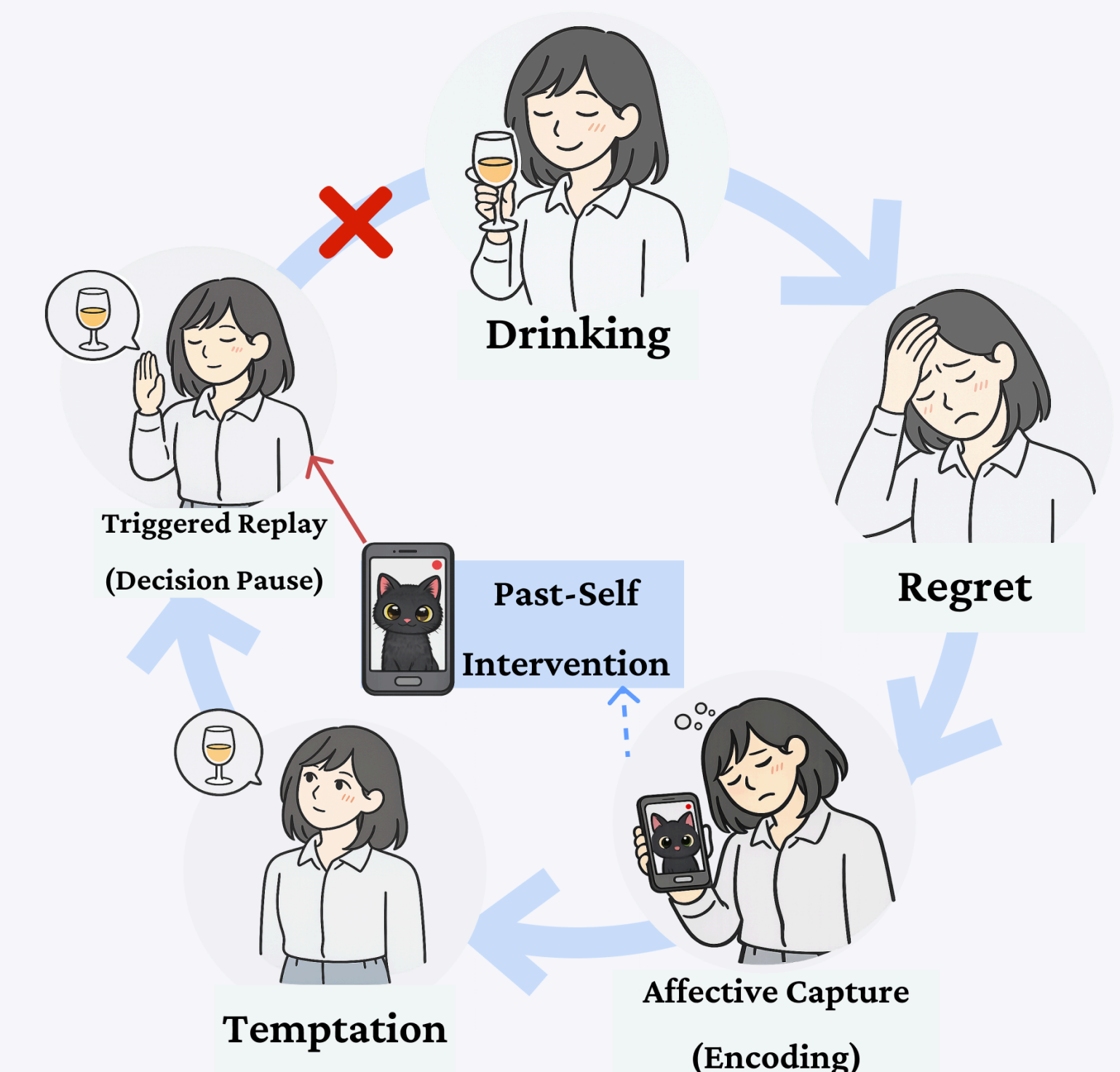


Figure 2. Comparison of affective-behavioral cycles with and without past-self intervention. Rewind introduces replay at temptation moments to interrupt the transition from temptation to action and create a decision pause.

Mechanism Overview

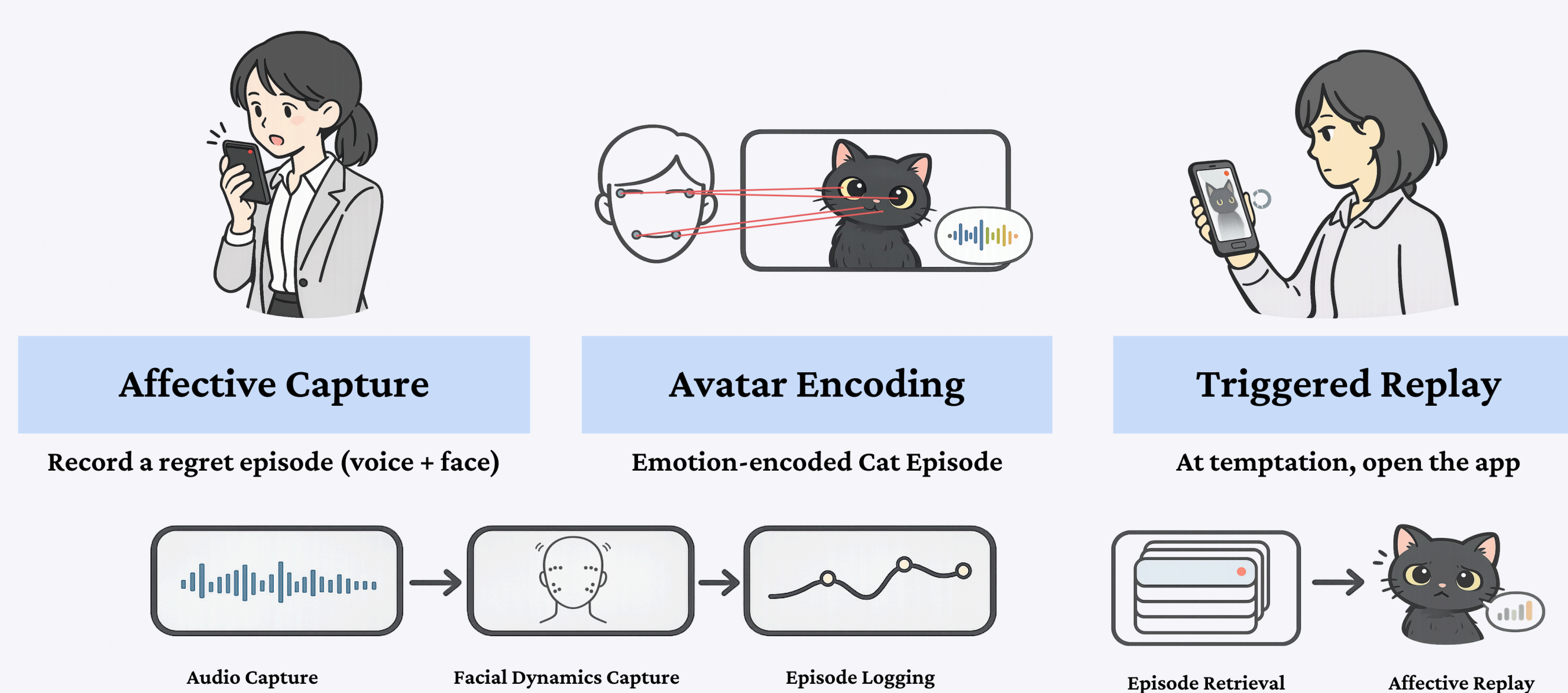


Figure 3. Mechanism overview of Rewind. A regret episode is captured, transformed into a replayable avatar-mediated artifact, and later replayed at temptation moments.

Methodology

Affective Capture

Record regret through voice and facial expression.

Avatar Encoding

Transform the episode into a condensed cat mediated replay artifact.

Triggered Replay

Replay the archived episode at later temptation moments to create a decision pause.

Motivation

Many unwanted habits do not happen because people lack awareness; they happen because awareness often comes too late. After a regretful moment, a person may sincerely promise to change, but when temptation returns, that emotional force often weakens. This pattern relates to the hot-cold empathy gap and future self-continuity, where short-term impulses can override earlier intentions when the future self feels psychologically distant [1,2].

Existing behavior-change systems often support planning, reflection, or restriction, but they still depend on willpower at the exact moment of temptation. Rewind is motivated by a simple question: could a system bring back how someone felt in the regretful moment, and use that replay to create a brief pause for reflection and reconsideration [3,4]?

Future Work

Future work will focus on building and refining the full Rewind prototype, including episode archiving, AI-assisted replay generation, and character-mediated playback. This next phase will move the concept toward a more complete and usable system.

It will also examine whether users initiate replay during real temptation moments, whether replay creates a measurable decision pause before action, and how repeated use may shape habit frequency over time. A further direction is to compare no-reward replay with rewarded replay, while exploring how replay can support reflection without feeling overly judgmental or intrusive.

References

- [1] Loewenstein, G. (2005). Hot-cold empathy gaps and medical decision making. *Health Psychology*.
- [2] Hershfield, H. E. (2011). Future self-continuity: How conceptions of the future self transform intertemporal choice. *Annals of the New York Academy of Sciences*, 1235(1), 30–43.
- [3] Nahum-Shani, I. (2018) Just-in-Time Adaptive Interventions (JITAs) in Mobile Health: Key Components and Design Principles for Ongoing Health Behavior Support.